

Experiment No. 5

Aim: Design a Login Screen Using UI Components and Event Handlers

Apparatus / Software Required :

Hardware	- Laptop(ASUS Vivobook 16x) - Minimum 8 GB RAM - AMD Ryzen 5 5600H with Radeon Graphics Processor - Internet Connection
Software	- Windows 11 Operating System - Android Studio (Quail) - Java Development Kit (JDK) - Android SDK - Android Emulator (Pixel 7) - Gradle Build System

Theory

A login screen is used to allow users to securely access an Android application. In Android, UI components such as TextView, EditText, ImageView and Button are used to design the login interface.

Event handlers are used to perform an action when the user interacts with the application.

For example, when the user clicks the Login button, the application checks the entered details and performs the required action.

In this experiment, a Student Portal login screen was designed. The screen contains fields for email/username and password, Login button, Forgot Password option, phone number login, and Sign Up option. Firebase Authentication is also used for authentication-related operations. The output shows successful login, registration, OTP login and password reset operations. The login screen and registration screen are shown on pages 5 and 6 of the uploaded file, while the successful login and phone login screens are shown on pages 7 and 8.

Implementation Steps:

1. Open **Android Studio**.
2. Create a new Android project.
3. Select **Java** as the programming language.
4. Configure the Android SDK and emulator.

5. Create a **Pixel 7** virtual device using **API 34**.
6. Open the `activity_main.xml` file.
7. Design the Student Portal login screen using different UI components.
8. Add an image/logo at the top of the screen.
9. Add an **Email or Username** input field.
10. Add a **Password** input field.
11. Add a **Forgot Password** option.
12. Add a **LOGIN** button.
13. Add a **Login with Phone Number** option.
14. Add a **Sign Up** option for new users.
15. Write Java code in `MainActivity.java` to handle user events.
16. Configure Firebase Authentication for authentication-related operations.
17. Test email/password login.
18. Test registration of a new user.
19. Test phone login and OTP verification.
20. Test the Forgot Password functionality.
21. Run the application on the Pixel 7 emulator and verify the output.

Files Used and Code Changes

1. `activity_main.xml`

This file is used to design the user interface of the login screen.

Changes made:

- Added Student Portal logo/image.
- Added Student Portal title.
- Added Email/Username EditText.
- Added Password EditText.
- Added Login Button.
- Added Forgot Password option.
- Added Phone Login button.
- Added Sign Up option.
- Added proper margins, padding and alignment.
- Applied background and text formatting.
- Used constraints to properly arrange the components.
- Added a second TextView displaying:

- “Developed by RAJ SIRSODE”
- Positioned the second TextView at the bottom center

2. MainActivity.java

This file is used to handle the events and operations of the login screen.

Changes made:

- Initialized the UI components.
- Added click events for buttons.
- Read the username/email and password entered by the user.
- Added login validation/authentication logic.
- Added registration event handling.
- Added Forgot Password event handling.
- Added phone login/OTP related event handling.
- Displayed appropriate messages such as Login Successful and OTP Sent.

3. Firebase Authentication

Firebase Authentication is used for managing authentication-related operations. The output in the uploaded file also shows the Firebase Authentication console containing registered users and authentication providers. Page 10 shows Firebase Authentication user records and a password-reset email received through Gmail.

Components Used

Component	Purpose
ConstraintLayout	Used to arrange the UI components
ImageView	Used to display the Student Portal/logo image
TextView	Used for headings, labels, Forgot Password and Sign Up
EditText	Used to enter email/username, password and OTP
Button	Used for Login, Register, Send OTP and Verify OTP operations
Toast	Used to display messages such as Login Successful and OTP Sent
Firebase Authentication	Used for user authentication and account management

Important XML Attributes Used

Attribute	Description
android:layout_width	Defines width of the view
android:layout_height	Defines height of the view
android:text	Sets the text displayed
android:textSize	Sets size of text
android:textStyle	Makes text bold
android:textColor	Sets text color
android:background	Sets background color
android:hint	Displays hint text inside an input field
android:inputType	Defines the type of input such as text, email or password
android:padding	Provides internal spacing
android:layout_margin	Provides external spacing
app:layout_constraintTop_toTopOf	Aligns a component with the top of another view
app:layout_constraintBottom_toBottomOf	Aligns a component with the bottom
app:layout_constraintStart_toStartOf	Aligns the start side of a component
app:layout_constraintEnd_toEndOf	Aligns the end side of a component

Important Java Methods Used

1. onCreate()

- It is called when the Activity is created. It is used to initialize the Activity and load the layout.
`protected void onCreate(Bundle savedInstanceState)`

2. setContentView()

- It connects the XML layout with the Java Activity.
`setContentView(R.layout.activity_main);`

3. findViewById()

- It is used to connect the UI components from XML with Java variables.
`EditText email = findViewById(R.id.email);`

4. **setOnClickListener()**

- It is used to handle the click event of buttons.

```
loginButton.setOnClickListener(v -> {  
    // Login operation  
});
```

5. **getText()**

- It is used to get the value entered by the user.

```
String email = emailEditText.getText().toString();
```

6. **Toast.makeText()**

- It is used to display short messages to the user.

```
Toast.makeText(this, "Login Successful", Toast.LENGTH_SHORT).show();
```

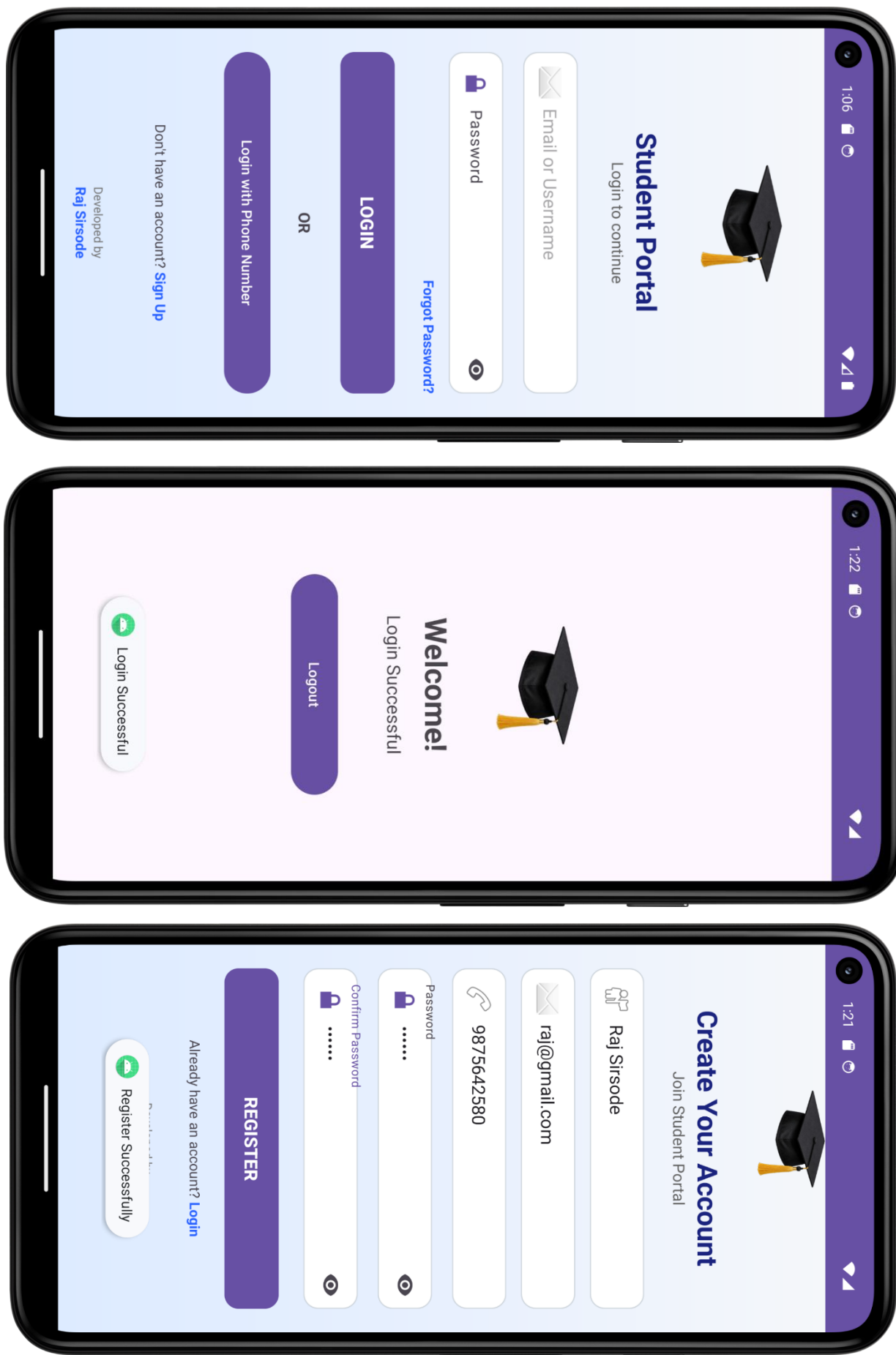
7. **Firebase Authentication Methods**

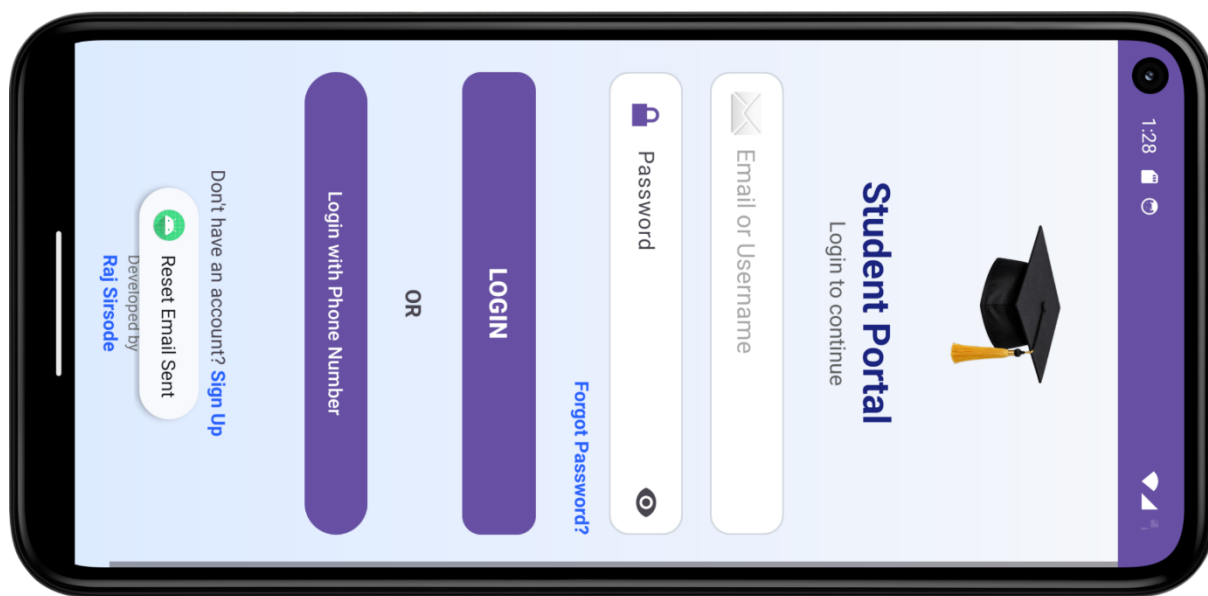
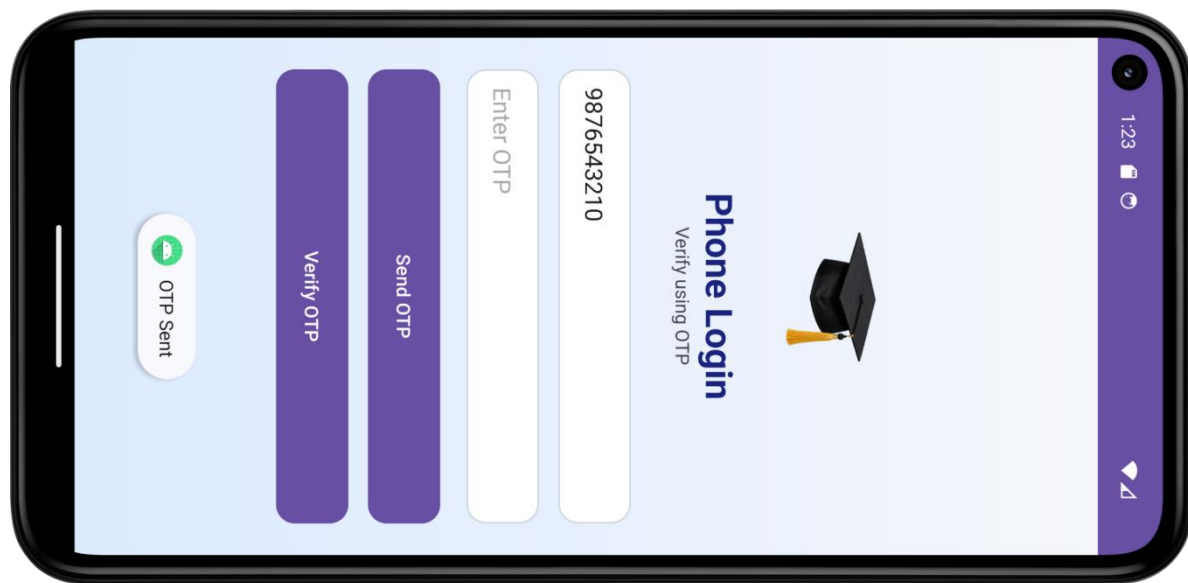
Firebase authentication methods are used for login, registration and password reset operations. The output also demonstrates phone/OTP authentication and password reset functionality.

Errors Faced and Troubleshooting

Error Faced	Troubleshooting
Gradle Sync Failed	Used compatible dependency versions and synchronized the project again
Incompatible Gradle JVM	Selected JVM 17
New library version error	Used compatible/downgraded dependency versions
Emulator setup issue	Created a Pixel 7 device with API 34
UI alignment problem	Added proper layout constraints
Login button not working	Checked the button ID and click listener
Invalid login details	Checked the entered username/email and password
OTP issue	Checked phone number, OTP field and authentication configuration
Password reset issue	Checked Firebase Authentication and reset-email configuration

Output:





Firebase Authentication ▾

Authentication

Users

Sign-in method

Templates

Usage

Settings

Extensions

The following authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

▾

🔍

Search by email address, phone number or user UID

Add user

↺

⋮

Identifier	Providers	Created ▾	Signed In	User UID
rajsirsode53@gmail.c...	✉	6 Aug 2026	6 Aug 2026	6OeyjYMIRWtUqRKLh3OcL2...
+919876543210	☎	6 Aug 2026	6 Aug 2026	jeR1cELtrCQKzM05Qf9DN1Q...

Rows per page

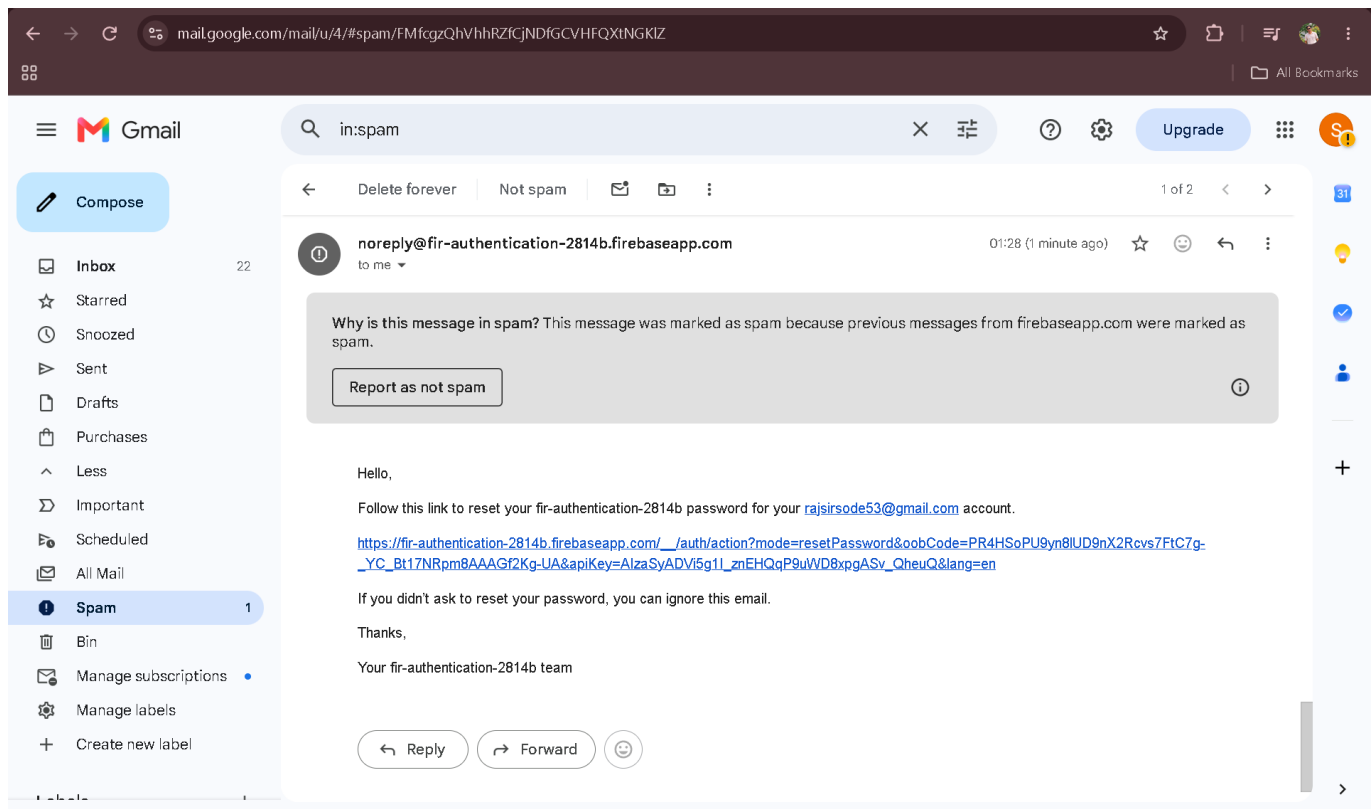
50 ▾

1 – 2 of 2

⏪

⏩

Mobile Application Development Lab | Experiment No. 5 | Raj Sirsode | Final Year-AIDS-B-104127



Conclusion

The Student Portal login screen was successfully designed using Android UI components and Java event handlers. Different components such as TextView, EditText, ImageView and Button were used to create the user interface. Event handlers were implemented to perform login, registration, phone/OTP login and password reset operations.

The application was successfully tested on the Pixel 7 emulator. The output demonstrated successful login, registration, OTP sending and password reset operations. Firebase Authentication was also used to manage authentication-related user information. Therefore, the experiment helped in understanding Android UI design, event handling and basic authentication functionality.

C (3)	A (3)	P (3)	T (9)	Sign